

Smart Cybersecurity for Emerging Technologies

Cybersecurity aims to avoid misuse and maintain confidentiality as well as the authenticity of electronic data by preventing unauthorised access via the exploitation of any loophole in cyber infrastructure.

We often come across various sign-up pages that tell us to set passwords while following certain basic rules, or verify that we are not a bot. Security is an essential element in the process of testing and validation of all hardware and software. For example, while having a conversation with others on a call, nobody wants a third party to listen to them. Yet, the list of most used passwords for encryption every year tells us that end-users often take the security of their accounts lightly until these are compromised.

With the growth of emerging technologies like the Internet of Things (IoT) and artificial intelligence (AI), and knowing the risk of cyber threats with traditional IT security tools, the importance of smart cybersecurity cannot be brushed under the rug.

Due to automation, modern systems are subject to robotic hackers too. An entity with malicious intent could control a robot or other device that works on commands through a man-in-the-middle attack. Cybersecurity aims to avoid misuse and maintain confidentiality as well as the authenticity of electronic data

by preventing unauthorised access via the exploitation of any loophole in cyber infrastructure. But without an upgrade, hackers could easily override common safeguards.

The answer to mitigating these challenges lies in leveraging these new technologies themselves. Data being at the epicentre of it all, it is necessary to categorise it. Often data is stored in a remote cloud environment. Having a robust and secure cloud infrastructure via proper authentication methods, accessing control, and using quantum cryptography can do wonders.



(Image source: www.mbtmag.com)

